

Andrea Tarditi

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EDUCATION (Politecnico di Milano)

MSc Mathematical Engineering, Quantitative Finance [exp. 110/110] 2023-Present

key courses: Financial Engineering, Mathematical Finance, Stochastic Calculus, Computational Finance, Algorithms and Parallel Computing, Fintech, Real and Functional Analysis, Insurance, Econometrics.

BSc Mathematical Engineering [94/110] 2020-2023

Mathematical Finance, Partial Differential Equations, Statistics, Numerical Analysis, Probability, Analysis I-III

Liceo Scientifico Statale R.Donatelli - B.Pascal 2015-2020

Student body president, focus on Mathematics, Physics, Philosophy, History

EXPERIENCE

Quantitative trader 2025-2026

Internship at UniCredit on the Government Bonds trading desk, where I support daily trading activities by developing and implementing quantitative tools and infrastructure.

Private Math & Physics teacher 2020-Present

Personalized tutoring in algebra, calculus and geometry for university and high school students.

Teacher Assistant 2018-2021

Assisted students with learning disabilities by creating personalized teaching strategies.

PROJECTS

[...see more on GitHub](#)

Multivariate Pricing 2024

Developed and calibrated multivariate Lévy models (NIG) for the S&P 500 and EURO STOXX 50; priced exotic derivatives and compared model performance against VG and Black models. [Matlab, Python]

Portfolio Replica 2025

Deployed a simulated fintech investment platform that uses machine-learning to classify investors and recommend tailored passive/active ETF portfolios. Built an interactive questionnaire for suitability prediction (e.g., Accumulation vs Income) and surfaced portfolio risk/TER details, with active models designed to flag market stress/anomalies.

Energy Price 2024

Modeled electricity futures using OU-IG and OU-Normal stochastic processes, capturing seasonality, jumps, and mean reversion; calibrated models to market data and simulated option prices via Montecarlo. [Matlab]

Insurance Company Capital Requirement 2024

Computed the Basic Solvency Capital Requirement for an insurance company based on Life and Market risks with both a stochastic and deterministic approach. [Matlab, Excel]

Clustering of Bank Customers 2025

Segmented bank's customers using clustering to identify key personas and optimize banking strategies. [Python]

Strategy Backtesting 2025

Used historical LOB simulations to derive optimal quoting policies, calibrating either the target spread from mid or the volume/position kept in front at each book level. Translated replay results into deployable rules linking quote aggressiveness to expected fills, adverse selection, and inventory risk.

Yield Curve decomposition 2025

Decomposed yield-curve dynamics via PCA into level (parallel shift), slope (twist), and curvature (butterfly) factors, separating Euro-wide components from country-specific movements.

Hawkes Market-Making 2026

Built and calibrated a bivariate Hawkes model of aggressive buy/sell order flow in limit order books. Formulated and solved an HJB stochastic-control market-making problem under CARA utility, modelling fills/adverse selection via predictable thinning/marking, and validated the resulting quoting policy in historical replay/backtests.

SKILLS & INTERESTS

Languages: Italian, English

Skills & tools: Matlab, C++, Python, Excel,

Latex, PowerPoint, R, Bloomberg terminal

AWARDS

Best Pitch Award 2024

QFinLab (Financial Engineering): winning pitch on multivariate Lévy models for exotic option pricing.